Complete-Graph Extremality under Death–Birth Updating: Fitness-Two Local Optimality and Strong-Selection Rigidity

Alec Kriebel
Independent Researcher
me@aleckriebel.com
ORCID 0009-0001-9320-500X

August 20, 2026

Abstract

We study fixation from a uniformly random single mutant under death–Birth updating on finite loopless directed weighted population structures. After normalizing each target's incoming weights, the dynamics are parametrized by a loopless row-stochastic replacement kernel; let J_n denote the uniform complete kernel. We prove two complementary extremality results. First, at mutant fitness two, J_n is a strict nondegenerate local maximizer in the full normalized kernel polytope for every $n \geq 3$. This includes arbitrary directed and nonreversible perturbations. The proof represents the fixation committor as a completely alternating coverage function of a fair-geometric union dual, converts fixation into a stationary collision observable, and decomposes the inverse-mean Hessian into standard, symmetric-balanced, and antisymmetric-balanced permutation sectors with all-order positive certificates. Second, no fixed finite directed weighting is a strict amplifier for every beneficial fitness. On complete support the leading strong-selection deficit is an explicit sum of squared discrepancies within incoming columns, with equality exactly for kernels dynamically equivalent to J_n; the remaining support cases follow from a source-component bound and a known noncomplete-support obstruction. We also prove global all-beneficial-fitness maximality on positive weighted triangles and two symmetric weighted K_4 families. These results establish full local rigidity at fitness two and fixed-graph rigidity at strong selection, but not global complete-graph maximality at fitness two.

Keywords. death–Birth updating; fixation probability; evolutionary graph theory; coverage duality; local optimality; strong selection.

Mathematics Subject Classification (2020). 92D15; 60J10; 05C81.

Author summary

Spatial structure can make a beneficial mutation more or less likely to take over a finite population. For death–Birth updating, the well-mixed population is widely suspected to be extremal, but the full statement remains open. We prove two complementary pieces of that picture. At fitness two, the well-mixed kernel is strictly locally optimal against every infinitesimal perturbation of its normalized replacement kernel, including directed and nonreversible ones. At the opposite, strong-selection end, every fixed finite structure is eventually suppressing unless its replacement dynamics are exactly well mixed. The local result comes from an exact ancestry process: fixation becomes a coverage probability, then a stationary collision rate, whose curvature splits into three symmetry types. The second result comes from a sum of squares measuring unequal competitors for the same death event. Weighted triangles provide a global bridge: every nonuniform positive triangle suppresses fixation at every beneficial fitness. The results do not show that the complete graph is globally best at fitness two or exclude graph sequences whose size threshold depends on fitness.

1 Introduction

The Moran process is a foundational finite-population model of selection and genetic drift (Moran, 1958). On a structured population, the fixation probability of a new mutant depends on both selection and the geometry of replacement. The complete graph is the well-mixed baseline, while directed and weighted evolutionary graphs can amplify or suppress selection under particular update rules and fitness ranges (Lieberman et al., 2005; Shakarian et al., 2012; Masuda and Ohtsuki, 2009; Allen et al., 2017, 2020). Birth–death and death–Birth updating behave especially differently (Kaveh et al., 2015): the latter exposes every mutant to a uniformly sampled death event before local competition, which severely constrains amplification (Hindersin and Traulsen, 2015; Tkadlec et al., 2020).

For a loopless directed weighting, multiply all incoming weights at a fixed target by the same positive number. This changes no death–Birth transition. The intrinsic object is therefore a loopless row-stochastic replacement kernel P, with a row indexed by the dead target and a column by a potential parent. Let J_n be the uniform off-diagonal kernel and write

$$D_n(P, r) = \rho_{\mathrm{dB}}(J_n, r) - \rho_{\mathrm{dB}}(P, r) \tag{1.1}$$

for the uniformly initialized fixation deficit. Global nonnegativity of $D_n(P, 2)$ for every n and every undirected weighted graph remains open. This paper resolves two transverse rigidity problems for the same surface $D_n(P, r)$.

First, we approach the complete kernel in structure space while holding fitness fixed at two. For every $n \geq 3$ and every nonzero loopless row-zero perturbation δ,

$$D_n(J_n + \varepsilon\delta, 2) = \kappa_n(\delta)\varepsilon^2 + O(\varepsilon^3), \qquad \kappa_n(\delta) > 0. \tag{1.2}$$

Thus J_n is a strict nondegenerate local maximizer among all positive loopless row-stochastic replacement kernels. This is stronger than a reversible or undirected statement: column imbalance and directed circulation are both allowed. It is nevertheless genuinely local. Its neighborhood may depend on n, and it does not constrain a sequence that stays far from J_n.

Second, we hold an arbitrary finite structure fixed and let $r \to \infty$. On complete directed support,

$$D_n(P, r) = \frac{\mathcal{E}_{\mathrm{dir}}(P)}{n^2(n-2)r} + O(r^{-2}), \tag{1.3}$$

where $\mathcal{E}_{\mathrm{dir}}$ is an explicit incoming-column sum of squares. Its zero set consists exactly of the kernels dynamically equivalent to J_n. Together with the noncomplete-support theorem of Tkadlec et al. (2020) and a source-component bound, this rules out one fixed finite directed weighting that amplifies for every $r > 1$.

The mechanism behind (1.2) is different from ordinary weak-selection perturbation. At $r = 2$, a death–Birth update is an additive Boolean OR of a fair-geometric number of potential parents. Reversing the graphical maps produces an exact finite union-valued ancestry chain; no independent-lineage approximation is made. Its stationary law represents the fixation committor as a completely alternating coverage function. A marked version turns the stationary mean ancestral-set size into a collision probability. At J_n, the first variation vanishes by rank projection. The Hessian is an

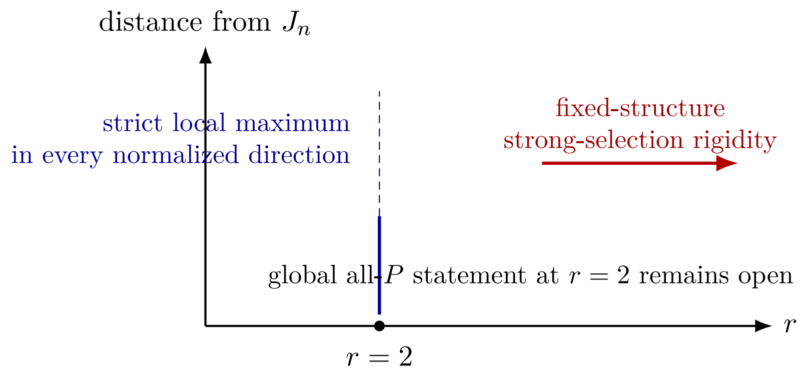

Figure 1: The two proved axes of the same complete-kernel deficit. The fitness-two theorem is local in the full normalized structure space and all-order in population size; the strong-selection theorem is global in fixed finite structure and asymptotic in fitness.

S_n-invariant quadratic form on all loopless row-zero matrices. The tangent space splits without multiplicity into standard, symmetric-balanced, and antisymmetric-balanced sectors. We prove all three scalars positive by rank-recursion and phase-contraction certificates. The standard sector is exactly the column-imbalance direction and requires a separate signed three-channel argument.

The global low-order results sharpen the picture. Every nonuniform positive weighted triangle is a strict suppressor for every $r > 1$. Two maximally symmetric weighted K_4 families obey the same conclusion. These theorems are not inferred from local curvature; they follow from independent exact sum-of-squares certificates.

An earlier unrefereed research release established the strong-selection and low-order portions of this argument (Kriebel, 2026). The present manuscript is a major superseding version. The coverage and collision representations, the full directed fitness-two Hessian theorem, the three all-order sector certificates, and the synthesis of the two extremal axes are new here; the earlier strong-selection and low-order proofs are retained and reorganized so that the quantifier boundaries can be compared in one place.

The quantifiers matter. The fixed-graph theorem proves

$$\forall P\ \exists r_0(P)\ \forall r > r_0(P): \quad D_n(P, r) > 0$$

unless P is dynamically complete. It does not rule out a single fitness-independent sequence (P_n) satisfying, for each fixed r, $D_n(P_n, r) < 0$ once n is sufficiently large. We keep this distinction explicit throughout.

The paper is organized as follows. Section 2 states the model and theorem suite. Sections 3–4 prove the fitness-two local theorem. Section 5 treats fixed structures at strong selection. Section 6 gives the global low-order classifications. The appendices contain the exact phase certificates, the symmetric K_4 algebra, and a quantifier ledger.

2 Model and main results

Let $V = \{1, \ldots, n\}$ and let $w_{uv} \geq 0$ be the weight with which source u competes to replace target v. We assume $w_{vv} = 0$ and every incoming degree $d_v^- = \sum_{u \neq v} w_{uv}$ is positive. Normalize

$$P_{vu} = \frac{w_{uv}}{d_v^-}. \tag{2.1}$$

Then P is loopless and row stochastic. Conversely, every such P is a normalized directed replacement structure. This quotient removes exactly the dynamically irrelevant positive rescalings of individual incoming columns.

A uniformly chosen target dies. If the current mutant set is S and mutant fitness is $r > 0$, the probability that target v receives a mutant parent is

$$f_r(P_{vS}) = \frac{rP_{vS}}{1+(r-1)P_{vS}}, \qquad P_{vS} = \sum_{u\in S} P_{vu}. \tag{2.2}$$

The all-resident and all-mutant states are absorbing. We write $\rho_{\mathrm{dB}}(P, r)$ for fixation probability averaged over a uniformly random singleton mutant. For the complete kernel

$$(J_n)_{vu} = \frac{1}{n-1} \quad (u \neq v), \tag{2.3}$$

one-dimensional first-step equations give

$$\rho_{\mathrm{dB}}(J_n, r) = \frac{n-1}{n}\frac{1-r^{-1}}{1-r^{-(n-1)}}. \tag{2.4}$$

At $r = 1$, the right-hand side is understood by continuity and equals $1/n$.

Let

$$\mathcal{T}_n = \{\delta \in \mathbb{R}^{n\times n} : \delta_{ii} = 0,\ \delta\mathbf{1} = 0\} \tag{2.5}$$

be the tangent space to the normalized kernel polytope at J_n, and let

$$\mathcal{B}_n = \{\delta \in \mathcal{T}_n : \mathbf{1}^{\mathsf{T}}\delta = 0\} \tag{2.5b}$$

be its balanced, or bistochastic, tangent subspace.

Theorem 1 (fitness-two strict local optimality)**.** *For every $n \geq 3$ there is a quadratic form $\mathcal{R}_n^{(2)}$ on $\mathcal{T}_n$ such that, for every $\delta \in \mathcal{T}_n$ and all sufficiently small ε for which $J_n + \varepsilon\delta$ is nonnegative,*

$$\frac{1}{n\rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)} = \frac{1}{m_n} + \varepsilon^2\mathcal{R}_n^{(2)}(\delta) + O(\varepsilon^3), \tag{2.6}$$

where

$$m_n = \frac{(n-1)2^{n-2}}{2^{n-1}-1}. \tag{2.7}$$

The quadratic form $\mathcal{R}_n^{(2)}$ is positive definite on $\mathcal{T}_n$. Consequently, the first variation vanishes for every $\delta \in \mathcal{T}_n$, and, for every nonzero $\delta \in \mathcal{T}_n$,

$$\left.\frac{d}{d\varepsilon}\rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)\right|_0 = 0, \qquad \left.\frac{d^2}{d\varepsilon^2}\rho_{\mathrm{dB}}(J_n + \varepsilon\delta, 2)\right|_0 = -\frac{2m_n^2}{n}\mathcal{R}_n^{(2)}(\delta) < 0. \tag{2.8}$$

Thus J_n is a strict nondegenerate local maximizer in the full positive loopless row-stochastic kernel polytope. For $n = 2$ every positive weighting induces the same two-vertex chain and ties the baseline.

For the fixed-structure result, return to unnormalized weights on complete directed support and define

$$\mathcal{E}_{\mathrm{dir}}(W) = \sum_{v}\sum_{\substack{u<z\\ u,z\neq v}} \frac{(w_{uv} - w_{zv})^2}{w_{uv}w_{zv}}. \tag{2.9}$$

Theorem 2 (complete-support strong-selection correction)**.** *Let $n \geq 3$ and suppose $w_{uv} > 0$ for every $u \neq v$. For fixed W,*

$$\rho_{\mathrm{dB}}(J_n, r) - \rho_{\mathrm{dB}}(W, r) = \frac{\mathcal{E}_{\mathrm{dir}}(W)}{n^2(n-2)r} + O(r^{-2}) \qquad (r \to \infty). \tag{2.10}$$

The defect vanishes if and only if, for every target v, all incoming weights w_{uv} with $u \neq v$ are equal. In that equality class the entire chain, not merely the asymptotic expansion, equals the complete-graph chain.

Corollary 3 (no fixed finite universal dB amplifier)**.** *No finite loopless directed weighting with positive incoming degrees is a strict dB amplifier for every $r > 1$. More precisely, it is strictly suppressing for all sufficiently large fitness unless it has complete support and is dynamically equivalent to J_n; in the latter case it ties the baseline for every fitness.*

For undirected weights, write s_i for the number of positive-weight neighbors of i.

Proposition 4 (undirected support limit)**.** *For every finite connected undirected weighted graph,*

$$\lim_{r\to\infty} \rho_{\mathrm{dB}}(W, r) = \frac{1}{n}\sum_{i=1}^{n} \frac{s_i}{s_i+1}. \tag{2.11}$$

Hence an incomplete support has the positive limiting deficit

$$\frac{n-1}{n} - \frac{1}{n}\sum_i \frac{s_i}{s_i+1} = \sum_i \frac{n-1-s_i}{n^2(s_i+1)}. \tag{2.12}$$

Finally, the complete kernel is globally extremal on the smallest nontrivial undirected order.

Theorem 5 (weighted triangles)**.** *For every positive undirected weighted triangle and every $r > 1$,*

$$\rho_{\mathrm{dB}}(W, r) \leq \rho_{\mathrm{dB}}(J_3, r) = \frac{2r}{3(r+1)}. \tag{2.13}$$

Equality holds if and only if the three edge weights are equal.

The two symmetric K_4 families and their equality statements are recorded in Appendix B. They provide global four-vertex slices but are not needed for Theorems 1 or 2.

3 Boolean ancestry, coverage, and collision

This section is valid for every positive loopless row-stochastic P; reversibility is not used.

3.1 The fair-geometric union dual

At fitness two, (2.2) becomes $2x/(1+x)$. This is a finite additive graphical system in the sense of Harris (1978). Let L have the fair geometric law

$$\mathbb{P}(L = \ell) = 2^{-\ell}, \qquad \ell \geq 1, \tag{3.1}$$

and, conditionally on target v, draw $U_1, \ldots, U_L$ independently from row P_v. For any mutant set S,

$$\mathbb{P}(U_1, \ldots, U_L \notin S) = \mathbb{E}(1 - P_{vS})^L = \frac{1 - P_{vS}}{1 + P_{vS}}. \tag{3.2}$$

Thus the forward update at v is exactly the additive Boolean map

$$x_v \longleftarrow x_{U_1} \vee \cdots \vee x_{U_L}. \tag{3.3}$$

The transpose graphical map sends a nonempty ancestral set A to

$$A \longmapsto \begin{cases} (A \setminus \{v\}) \cup \{U_1, \dots, U_L\}, & v \in A, \\ A, & v \notin A. \end{cases} \tag{3.4}$$

Because $P_{vv} = 0$, the updated set in the first line omits v and is always nonempty; hence the dual remains a proper nonempty subset whenever it starts there.

Using the same graphical marks forward and backward gives the exact duality

$$\mathbb{P}_S(X_t \cap A \neq \varnothing) = \mathbb{P}_A(S \cap A_t \neq \varnothing). \tag{3.5}$$

For positive P, the dual is irreducible on the proper nonempty subsets. Let Π_P denote its stationary law and put

$$m(P) = \mathbb{E}_{\Pi_P}|A|. \tag{3.6}$$

Proposition 6 (coverage representation)**.** *The fixation committor at fitness two is*

$$h_P(S) = \mathbb{P}_{A\sim\Pi_P}(A \cap S \neq \varnothing), \tag{3.7}$$

and therefore

$$\rho_{\mathrm{dB}}(P, 2) = \frac{m(P)}{n}. \tag{3.8}$$

Moreover, for every nonempty T disjoint from S,

$$(-1)^{|T|+1}\Delta_T h_P(S) = \mathbb{P}_{\Pi_P}(A \cap S = \varnothing,\ T \subseteq A) \geq 0. \tag{3.9}$$

Thus h_P is a normalized completely alternating coverage function.

Proof. The finite forward chain absorbs almost surely. Start the dual from V in (3.5) and let $t \to \infty$. The left side converges to the fixation probability from S, while the recurrent dual law converges to Π_P, giving (3.7). Averaging $h_P(\{i\}) = \mathbb{P}(i \in A)$ over i proves (3.8). For fixed A, the indicator $\mathbf{1}_{\{A\cap S\neq\varnothing\}}$ has mixed difference $(-1)^{|T|+1}\Delta_T$ equal to $\mathbf{1}_{\{A\cap S=\varnothing,\ T\subseteq A\}}$; averaging proves (3.9). □

3.2 A one-sample active chain

The geometric burst can be factored into one-sample steps. Let

$$\mathcal{Y}_n = \{(B, v) : \varnothing \neq B \subseteq V \setminus \{v\}\}. \tag{3.10}$$

From $y = (B, v)$, with $b = |B|$, define $K(P)$ by the following fair mixture:

1. sample i from row P_v and move to $(B \cup \{i\}, v)$;
2. choose w uniformly from B, sample i from row P_w, and move to $((B \setminus \{w\}) \cup \{i\}, w)$.

Each branch has probability 1/2. The first branch continues a geometric burst; the second stops it and retargets the active ancestral mark.

Lemma 7 (active collision identity)**.** *Let $\nu(P)$ be the stationary law of $K(P)$ and put*

$$H(B,v)=\frac{1}{|B|}. \tag{3.11}$$

Then

$$\nu(P)H=\frac{1}{m(P)}=\frac{1}{n\rho_{\mathrm{dB}}(P,2)}. \tag{3.12}$$

The map $P\mapsto K(P)$ is linear.

Proof. We include the marked-current calculation because it is the bridge from the coverage law to the Hessian. Let $T_v(A,B)$ be the transition kernel of one union-dual update with target fixed at v. For $v\notin C$, define

$$\sigma_v(C)=\Pi_P(C\cup\{v\}),\qquad \eta_v(C)=\sum_A \Pi_P(A)T_v(A,C)-\Pi_P(C). \tag{3.13a}$$

The subtracted term is exactly the passive current from $A=C$, for which $v\notin A$ and no ancestral update occurs; hence $\eta_v(C)$ is the nonnegative active current arriving at output (C,v). Stationarity of the union dual, after summing its n equally likely target updates, gives

$$\sum_{v\notin B}\eta_v(B)=|B|\Pi_P(B). \tag{3.13}$$

If A_v adjoins one sample from row P_v, the fair-geometric resolvent identity gives

$$\lambda_v=\frac{\sigma_v+\eta_v}{2},\qquad \eta_v=\lambda_v A_v. \tag{3.14}$$

Write A for the one-sample channel and R for the fair continue-or-retarget channel. After the sample, continuing contributes $\eta_v(B)/2$ to (B,v). Stopping, choosing w uniformly from B, and ending at $(B\setminus\{w\},w)$ contributes, by (3.13), $\Pi_P(B)/2=\sigma_w(B\setminus\{w\})/2$. Hence $\lambda AR=\lambda$. Since $\eta=\lambda A$, associativity gives $\eta RA=\eta$; the push-forward RA is precisely the active kernel $K(P)$ defined above. Both σ_v and η_v have total mass $\mathbb{P}_{\Pi_P}(v\in A)$, so

$$\sum_{v,C}\lambda_v(C)=m(P). \tag{3.15}$$

Normalizing the pushed-forward law gives $\nu(P)=\eta/m(P)$. Dividing (3.13) by $|B|$ and summing over B yields

$$\nu(P)H=\sum_{k\geq 1}\frac{k\Pi_P(|A|=k)}{m(P)}\frac{1}{k}=\frac{1}{m(P)}. \tag{3.16}$$

Both active moves contain exactly one use of a row of P, proving linearity. □

At the complete kernel, put $N=n-1$. Direct substitution gives

$$\nu_0(B,v)=\frac{|B|}{nN2^{N-1}},\qquad c_0:=\nu_0H=\frac{2^N-1}{N2^{N-1}}=\frac{1}{m_n}. \tag{3.17}$$

This agrees with (2.4) at $r=2$.

The collision interpretation is literal. In one stationary marked-chain step, let I be the sampled source; if the fair continue-or-stop coin stops, let W be the new target chosen uniformly from B. Then

$$\mathbb{P}(\text{stop},W=I)=\frac{1}{2m(P)},\qquad \mathbb{P}(W=I\mid\text{stop})=\frac{1}{m(P)}. \tag{3.18}$$

Thus maximizing fixation is equivalent to minimizing this stationary collision rate.

4 The complete-kernel Hessian

Let $P_\varepsilon = J_n + \varepsilon\delta$ with $\delta \in \mathcal{T}_n$. By linearity,

$$K(P_\varepsilon) = K_0 + \varepsilon\Delta. \tag{4.1}$$

All entries are analytic for ε near zero, and K_0 is irreducible. Put

$$q = H - c_0\mathbf{1}, \qquad G = (I - K_0 + \mathbf{1}\nu_0)^{-1}. \tag{4.2}$$

4.1 Vanishing first variation

Let S average a function on $\mathcal{Y}_n$ over all active states of the same rank $|B|$. Permutation symmetry gives $SK_0 = K_0S$. A direct average of either active move shows

$$S\Delta S = 0. \tag{4.3}$$

Indeed, a uniformly random k-subset of $V \setminus \{v\}$ contains a sample from any row with mean probability k/N; after retargeting the corresponding mean is $(k-1)/N$. More explicitly, if f_k is any rank function, the two branchwise perturbations at (B, v) are

$$\frac{f_{k+1} - f_k}{2} \sum_{i \notin B} \delta_{vi}, \qquad \frac{f_{k-1} - f_k}{2k} \sum_{w \in B} \sum_{i \in B \setminus \{w\}} \delta_{wi}. \tag{4.3a}$$

Under the uniform rank-k orbit their respective row sums are

$$\frac{N-k}{N} \sum_{i \neq v} \delta_{vi} = 0, \qquad \frac{k-1}{N} \sum_{i \neq w} \delta_{wi} = 0. \tag{4.3b}$$

This proves (4.3) without reversibility. Because q and Gq are rank functions and $\nu_0 = \nu_0 S$,

$$\nu_0 \Delta G q = \nu_0 S \Delta S G q = 0. \tag{4.4}$$

Stationary perturbation of the finite chain gives

$$\nu(P_\varepsilon) q = \varepsilon \nu_0 \Delta G q + \varepsilon^2 \nu_0 \Delta G \Delta G q + O(\varepsilon^3). \tag{4.5}$$

Define

$$\mathcal{R}_n^{(2)}(\delta) = \nu_0 \Delta G \Delta G q. \tag{4.6}$$

Equations (3.12), (3.17), and (4.4)–(4.6) already prove expansion (2.6); it remains to prove that $\mathcal{R}_n^{(2)}$ is positive definite on the full tangent space.

4.2 The three irreducible tangent sectors

For $\delta \in \mathcal{T}_n$, let

$$s_j = \sum_i \delta_{ij}, \qquad \sum_j s_j = 0, \tag{4.7}$$

and define the standard embedding

$$E(s)_{ij} = \frac{s_i + (n-1)s_j}{n(n-2)} \quad (i \neq j), \qquad E(s)_{ii} = 0. \tag{4.8}$$

Then $E(s)\mathbf{1} = 0$ and its column-sum vector is s. Consequently $B = \delta - E(s)$ has zero row and column sums, and

$$\delta = E(s) + \frac{B + B^{\mathsf{T}}}{2} + \frac{B - B^{\mathsf{T}}}{2}. \tag{4.9}$$

The three terms are Frobenius-orthogonal. Their dimensions are

$$n - 1, \qquad \frac{n(n-3)}{2}, \qquad \frac{(n-1)(n-2)}{2}, \tag{4.10}$$

which sum to $n(n-2) = \dim \mathcal{T}_n$. They are, respectively, the standard, symmetric row/column-balanced, and antisymmetric balanced irreducible S_n-modules. At $n = 3$ the symmetric module has dimension zero.

For completeness, the symmetric off-diagonal matrices form the two-subset permutation module $\mathbf{1} \oplus [n-1, 1] \oplus [n-2, 2]$; its row-balanced kernel is $[n-2, 2]$. The antisymmetric balanced kernel is $\bigwedge^2 [n-1, 1] = [n-2, 1, 1]$, while $E(s)$ realizes $[n-1, 1]$. Thus the three summands in (4.9) are pairwise nonisomorphic and occur with multiplicity one, including the stated $n = 3$ degeneration.

The quadratic form (4.6) is invariant under simultaneous relabeling of targets and sources. Multiplicity-freeness therefore makes the three spaces orthogonal for $\mathcal{R}_n^{(2)}$ as well, and $\mathcal{R}_n^{(2)}$ acts by one scalar on each. The balanced subspace is exactly

$$\mathcal{B}_n = \mathcal{S}_n \oplus \mathcal{A}_n, \tag{4.10b}$$

where $\mathcal{S}_n$ and $\mathcal{A}_n$ denote the symmetric- and antisymmetric-balanced summands in (4.9). The standard module $E(s)$ is precisely the complementary column-imbalance direction.

Theorem 8 (sector positivity)**.** *The standard scalar is positive for every $n \geq 3$, the symmetric-balanced scalar is positive for every $n \geq 4$, and the antisymmetric-balanced scalar is positive for every $n \geq 3$.*

The proof is in Appendix A. We summarize its architecture. For the standard sector, a signed three-channel quotient separates positive and negative orientations. Schur elimination turns each completed negative excursion into a nonnegative re-entry operator. Exact first-phase barriers and a uniform row-sum contraction control the entire alternating resolvent. For the antisymmetric sector, a one-feature reduction sends a rank Poisson gradient through a positive decreasing recurrence; the second perturbation is then a sum of nonnegative sampling-without-replacement terms, at least one strict. For the symmetric sector, the exact quotient has a good/bad block form

$$\begin{pmatrix} S & C \\ -D & Q \end{pmatrix} \tag{4.11}$$

with S, C, D, Q nonnegative. Schur elimination converts every completed bad excursion into one factor of a nonnegative return operator A. Explicit positive phase barriers dominate the alternating resolvent $(I + A)^{-1}$. All large orders are covered by coefficientwise polynomial certificates; the finitely many boundary orders are solved exactly over $\mathbb{Q}$ by the included replay.

For orientation, the first sector eigenvalues are

n	λ_{std}	λ_{sym}	λ_{anti}
3	1/11	$---$	1/9
4	87/640	3/208	57/640
5	8585/57314	359/26660	143/2100.

(4.12)

These are the Frobenius-normalized quadratic eigenvalues; they are checks, not a finite extrapolation.

Proof of Theorem 1. Theorem 8 and the decomposition (4.9) imply $\mathcal{R}_n^{(2)}(\delta) > 0$ for every nonzero $\delta \in \mathcal{T}_n$. Equations (3.12) and (4.5) give

$$\frac{1}{n\rho_{\mathrm{dB}}(P_\varepsilon, 2)} = c_0 + \varepsilon^2 \mathcal{R}_n^{(2)}(\delta) + O(\varepsilon^3). \tag{4.13}$$

Differentiation yields (2.8). To obtain a neighborhood rather than only directional statements, restrict the analytic fixation map to a small ball around J_n in the affine space $J_n + \mathcal{T}_n$. The smallest $\mathcal{R}_n^{(2)}$-eigenvalue on the unit sphere is positive. Taylor's theorem with a uniform third-derivative bound on a smaller compact ball makes the negative quadratic term dominate the remainder. Hence every other kernel in that ball has strictly smaller fixation probability. □

Remark 9. *The neighborhood in the last paragraph depends on n. No statement here prevents a population sequence from approaching a singular face, or from remaining a nonperturbative distance from J_n as $n \to \infty$.*

5 Fixed structures at strong selection

We now prove Theorem 2, Corollary 3, and Proposition 4.

5.1 Complete-graph baseline

Let ϕ_k be complete-graph fixation from k mutants and let γ_k be the ratio of the probabilities of decreasing and increasing k. First-step equations give

$$\phi_1 = \left(1 + \sum_{j=1}^{n-1} \prod_{k=1}^{j} \gamma_k\right)^{-1}. \tag{5.1}$$

Under dB updating,

$$\gamma_k = \frac{n-1+(r-1)k}{r\{n-1+(r-1)(k-1)\}}, \tag{5.2}$$

so the product telescopes and yields (2.4). For $n \geq 3$,

$$\rho_{\mathrm{dB}}(J_n, r) = \frac{n-1}{n} - \frac{n-1}{nr} + O(r^{-2}). \tag{5.3}$$

We use the following standard finite-state observation.

Lemma 10 (finite-state perturbation)**.** *Let a finite absorbing transition matrix depend analytically on ε near zero. If the transient restriction at $\varepsilon = 0$ has spectral radius below one, then all absorption probabilities are analytic near zero. More generally, if a limiting reachable transient class has a bounded fundamental matrix and one-step leakage is $O(\varepsilon)$, then leakage before absorption is $O(\varepsilon)$.*

Proof. Write the transient absorption equation as $(I - Q(\varepsilon))h = b$. The inverse exists near zero and is analytic by the adjugate formula. On a restricted limiting class, its bounded fundamental matrix times the leakage vector bounds the probability of leaving before absorption. □

5.2 Complete directed support

Assume $n \geq 3$ and $w_{uv} > 0$ for $u \neq v$. Put

$$\varepsilon = r^{-1}, \qquad a_{uv} = \frac{d_v^- - w_{uv}}{w_{uv}}. \tag{5.4}$$

Let $q_S(\varepsilon)$ be extinction probability from mutant set S. At $\varepsilon = 0$, every set with at least two mutants fixates, while $q_{\{i\}}(0) = 1/n$. The limiting transient chain absorbs, so Lemma 10 gives analyticity.

Differentiate the exact first-step system. If $3 \leq |S| \leq n-1$, a first-order loss still leaves at least two mutants, whose zeroth-order extinction is zero. The derivative therefore satisfies

$$(n - |S|)q'_S(0) = \sum_{v \notin S} q'_{S \cup \{v\}}(0). \tag{5.5}$$

Backward induction from $q'_V(0) = 0$ gives

$$q_S(\varepsilon) = O(\varepsilon^2), \qquad |S| \geq 3. \tag{5.6}$$

For a pair $S = \{i, j\}$, loss of target i has probability

$$\frac{1}{n} \frac{\varepsilon(d_i^- - w_{ji})}{w_{ji} + \varepsilon(d_i^- - w_{ji})} = \frac{\varepsilon a_{ji}}{n} + O(\varepsilon^2), \tag{5.7}$$

and similarly for target j. The leading self-loop mass is $2/n$, the leading gain mass is $(n-2)/n$, and a resulting singleton has extinction $1/n + O(\varepsilon)$. Hence

$$q_{\{i,j\}}(\varepsilon) = \frac{a_{ij} + a_{ji}}{n(n-2)} \varepsilon + O(\varepsilon^2). \tag{5.8}$$

From singleton $\{i\}$, death of resident target v creates the pair $\{i, v\}$ with probability

$$t_{iv} = \frac{1}{n} \frac{w_{iv}}{w_{iv} + \varepsilon(d_v^- - w_{iv})} = \frac{1}{n}(1 - a_{iv}\varepsilon) + O(\varepsilon^2). \tag{5.9}$$

Death of i causes extinction with probability $1/n$. Removing holding probabilities gives

$$\left(\frac{1}{n} + \sum_{v \neq i} t_{iv} \right) q_{\{i\}} = \frac{1}{n} + \sum_{v \neq i} t_{iv} q_{\{i,v\}}. \tag{5.10}$$

Define

$$O_i = \sum_{v \neq i} a_{iv}, \qquad I_i = \sum_{u \neq i} a_{ui}. \tag{5.11}$$

Substitution of (5.8) into (5.10) yields

$$q_{\{i\}}(\varepsilon) = \frac{1}{n} + \frac{\varepsilon}{n^2} \left[O_i + \frac{O_i + I_i}{n-2} \right] + O(\varepsilon^2). \tag{5.12}$$

Since $\sum_i O_i = \sum_i I_i =: T_{\mathrm{dir}}(W)$, uniform averaging gives

$$\rho_{\mathrm{dB}}(W, r) = \frac{n-1}{n} - \frac{T_{\mathrm{dir}}(W)}{n^2(n-2)r} + O(r^{-2}), \tag{5.13}$$

where

$$T_{\mathrm{dir}}(W) = \sum_v \sum_{u \neq v} \frac{d_v^- - w_{uv}}{w_{uv}}. \tag{5.14}$$

For each target column, the elementary identity

$$d_v^- \sum_{u \neq v} \frac{1}{w_{uv}} - (n-1)^2 = \sum_{\substack{u<z \\ u,z \neq v}} \frac{(w_{uv} - w_{zv})^2}{w_{uv} w_{zv}} \tag{5.15}$$

gives

$$T_{\mathrm{dir}}(W) - n(n-1)(n-2) = \mathcal{E}_{\mathrm{dir}}(W). \tag{5.16}$$

Equations (5.3), (5.13), and (5.16) prove Theorem 2. Equality in (5.15) forces each incoming column to be constant. Such column constants cancel from every parent competition, so the full chain is exactly the J_n chain.

5.3 Noncomplete supports and fixed-graph closure

If the positive directed support is strongly connected but noncomplete, Tkadlec et al. (2020) prove eventual strict dB suppression. It remains to remove strong connectivity.

Consider the condensation DAG. If it has at least two source strongly connected components, then a uniformly initialized singleton leaves at least one source component entirely resident and unable to receive mutant ancestry; fixation is impossible. Otherwise let $C \subsetneq V$ be the unique source component. Initial mutants outside C cannot fixate. For $i \in C$, let s_i^+ be its positive out-support size. While i is the only mutant, its chance to make a first gain before dying is at most $s_i^+/(s_i^+ + 1)$. Thus

$$\rho_{\mathrm{dB}}(W, r) \leq \frac{1}{n} \sum_{i \in C} \frac{s_i^+}{s_i^+ + 1} \leq \frac{(n-1)^2}{n^2}. \tag{5.17}$$

The complete baseline tends to $(n-1)/n$, leaving limiting gap $(n-1)/n^2$. Combining this case, the cited strongly connected noncomplete-support theorem, and Theorem 2 proves Corollary 3.

5.4 Undirected incomplete support

Fix an initial mutant i in a connected undirected support. At $r = \infty$, death of i causes extinction, while death of any one of its s_i neighbors creates an adjacent mutant pair. After deleting holding steps, the probability of reaching a pair first is $s_i/(s_i + 1)$. Once an adjacent pair exists, the mutant set remains support-connected; every mutant has a mutant neighbor and cannot be replaced by a resident in the limiting chain, while every boundary resident death produces monotone growth. The limiting chain therefore reaches fixation almost surely from the pair. Lemma 10 controls $O(1/r)$ leakage from this limiting class. Uniform averaging proves (2.11); elementary subtraction proves (2.12).

6 Global all-fitness slices at small order

Local curvature at $r = 2$ does not imply global maximality. At order three, however, the full all-fitness comparison admits an exact centered sum-of-squares certificate.

Let an undirected triangle have positive edge weights (a, b, c). If F_i is fixation when i is the unique mutant and G_i when i is the unique resident, deleting holding probabilities gives six linear equations. For distinct i, j, k, put

$$u_{ij} = \frac{rw_{ij}}{rw_{ij} + w_{jk}}, \qquad v_{ij} = \frac{w_{ij}}{rw_{jk} + w_{ij}}. \tag{6.1}$$

Then

$$(1 + u_{ij} + u_{ik})F_i = u_{ij}G_k + u_{ik}G_j, \tag{6.2}$$

$$(1 + v_{ij} + v_{ik})G_i = 1 + v_{ij}F_k + v_{ik}F_j. \tag{6.3}$$

The coefficient matrix is a strictly diagonally dominant M-matrix, so its determinant is positive.

Put $s_1 = a + b + c$, $s_2 = ab + ac + bc$, and $s_3 = abc$, and define

$$\begin{aligned} A &= 3s_3(s_1s_2 - 9s_3), \\ D &= 12s_1^3s_3 - 45s_1s_2s_3 + 4s_2^3 - 27s_3^2, \\ E &= 4s_2(3s_1^2s_2 - 3s_1s_3 - 8s_2^2), \\ \mathcal{H}_r &= A(r-1)^4 + Dr(r-1)^2 + Er^2. \end{aligned} \tag{6.4}$$

Exact elimination of (6.2)–(6.3) gives

$$\rho_{\mathrm{dB}}(W, r) - \rho_{\mathrm{dB}}(J_3, r) = -\frac{r(r-1)\mathcal{H}_r}{3(r+1)\mathcal{P}_r}, \tag{6.5}$$

where $\mathcal{P}_r > 0$ is the cleared determinant. More explicitly, if M is the coefficient matrix of (6.2)–(6.3) and

$$L = \prod_{\substack{p,q \in \{a,b,c\} \\ p \neq q}} (rp + q) > 0,$$

then $\mathcal{P}_r = L \det(M)/3 > 0$.

To sign the numerator, set

$$U = s_1^2 - 3s_2, \quad V = s_2^2 - 3s_1s_3, \quad W = s_1s_2 - 9s_3, \quad Z = s_2^3 - 27s_3^2. \tag{6.6}$$

Writing $X = ab$, $Y = ac$, $Z_0 = bc$ gives

$$\begin{aligned} 2U &= (a-b)^2 + (a-c)^2 + (b-c)^2, \\ 2V &= (X-Y)^2 + (X-Z_0)^2 + (Y-Z_0)^2, \\ W &= c(a-b)^2 + b(a-c)^2 + a(b-c)^2, \\ Z &= s_2V + 3\{Z_0(X-Y)^2 + Y(X-Z_0)^2 + X(Y-Z_0)^2\}. \end{aligned} \tag{6.7}$$

The coefficient identities

$$A = 3s_3W, \qquad D = 12s_1s_3U + 3s_2V + Z, \qquad E = 4s_2(3s_2U + V) \tag{6.8}$$

show that $A, D, E \geq 0$. If the weights are nonuniform then $U > 0$, hence $E > 0$ and $\mathcal{H}_r > 0$ for $r > 1$. Equation (6.5) proves Theorem 5, including its equality class.

Appendix B gives analogous exact certificates for the $1+3$ and $2+2$ invariant positive weighted K_4 families. Those families are included because they test the first genuinely higher-dimensional subset chains while retaining exact orbit reductions. We make no claim that they classify all weighted four-vertex graphs.

7 Implications, limitations, and reproducibility

7.1 A necessary condition for asymptotic universal amplification

The fixed-graph result does not interchange n and r. It nevertheless implies a useful support condition for undirected graph sequences. Fixation is monotone in fitness under a common dead target and common uniform update threshold. Proposition 4 therefore gives, for every finite connected undirected graph,

$$\rho_{\mathrm{dB}}(W, r) \le 1 - \frac{1}{n} \sum_i \frac{1}{s_i + 1}. \tag{7.1}$$

Proposition 11 (support degree must diverge)**.** *Let W_n be connected undirected weighted graphs of order $n \to \infty$. If, for every fixed $r > 1$, W_n is eventually a dB amplifier, then*

$$\frac{1}{n} \sum_i \frac{1}{s_i + 1} \longrightarrow 0. \tag{7.2}$$

Consequently the support degree of a uniformly chosen vertex tends to infinity in probability.

Proof. At fixed $R > 1$, eventual amplification and (7.1) give

$$\limsup_{n\to\infty} \frac{1}{n} \sum_i \frac{1}{s_i + 1} \le \lim_{n\to\infty} \{1 - \rho_{\mathrm{dB}}(J_n, R)\} = \frac{1}{R}.$$

Let $R \to \infty$. For fixed K, the fraction with $s_i \le K$ is at most $(K+1)$ times the average in (7.2). $\square$

This excludes uniformly bounded-support families, but not supports completed by arbitrarily weak positive edges, diffuse perturbations, mesoscopic modules, or singular population-dependent kernels.

7.2 What the local theorem does and does not say

Theorem 1 proves strict negative fixation curvature in every nonzero normalized directed direction at fixed order. In particular, neither column imbalance nor directed circulation escapes the local maximum. The theorem does not supply a radius uniform in n, global concavity along rays, or a path from an arbitrary kernel to J_n on which fixation is monotone. The global conjecture

$$\rho_{\mathrm{dB}}(P, 2) \le \rho_{\mathrm{dB}}(J_n, 2) \tag{7.3}$$

for every undirected weighted graph remains open.

7.3 Exact verification and proof status

The archive contains independent exact implementations for the two theorem axes. The strong-selection suite constructs literal subset-state chains, checks row normalization and complete-graph lumpability, differentiates the absorption equations, verifies the incoming-column sum of squares, and replays the triangle and K_4 certificates. The fitness-two suite independently builds the marked and active chains, verifies the collision normalization, reconstructs the rank Poisson equation, checks the tangent decomposition, and replays all three irreducible-sector certificates.

The standard sector has an analytic tail certificate for $N \ge 10$ and exact rational closure for $2 \le N \le 9$. The symmetric sector uses exact solves for $3 \le N \le 39$, exact rational phase margins for $40 \le N \le 287$, and a coefficient/discriminant certificate for $N \ge 288$. All arithmetic in these finite ranges is rational. The verifiers discharge the finite computer-assisted portions; sampled numerical values are never used to extend a quantifier.

The one-command replay is

```
./replay.sh
```

from the paper package. The package-level integration audit separately checks the complete active stationary law, the full tangent decomposition, all three physical sector normalizations, the strong-selection column identity, and conversion from inverse-mean to fixation curvature.

Statements and Declarations

Data and code availability. No empirical data are used. Complete LaTeX source, exact rational and symbolic verifiers, coefficient certificates, build scripts, and adversarial audit reports accompany this manuscript in the development repository https://github.com/AlecKriebel/Math. The earlier strong-selection and low-order version is archived at https://doi.org/10.5281/zenodo.21753405; that record does not contain the new fitness-two local theorem. The accompanying standalone source bundle freezes every dependency needed to replay the expanded manuscript and is intended for versioned deposit with the preprint.

Funding. No external funding supported this work.

Competing interests. The author declares no competing interests.

Author contributions. A.K. conceived the study, developed and checked the proofs, wrote and audited the exact software, and prepared the manuscript.

Ethics approval and consent. Not applicable. This study uses no human participants, animals, or empirical biological data.

AI-assisted research disclosure. Generative-AI systems were used substantively in mathematical exploration, derivation, adversarial proof analysis, software development, and manuscript preparation. The author determined the scope and claims, independently replayed the exact certificates, and accepts responsibility for the manuscript and accompanying materials.

A All-order Hessian-sector certificates

This appendix proves Theorem 8. Throughout, $N = n - 1$ and

$$\pi_k = \frac{\binom{N-1}{k-1}}{2^{N-1}}, \qquad 1 \le k \le N, \tag{A.1}$$

is the complete active-rank law. Its rank chain has

$$R_{k,k+1} = \frac{N-k}{2N}, \qquad R_{k,k-1} = \frac{k-1}{2N}. \tag{A.2}$$

Let h solve $(I - R)h = k^{-1} - c_0$ modulo constants and put $d_k = h_k - h_{k+1}$ for $1 \le k < N$, with $d_0 = d_N = 0$. Subtracting adjacent Poisson equations gives

$$(N-k)d_k - (k-1)d_{k-1} = 2N\left(\frac{1}{k} - c_0\right), \qquad 1 \le k < N. \tag{A.3}$$

Solving this recurrence gives, for $1 \le k < N$,

$$\frac{2}{k} - d_k = \frac{2^{2-N}\sum_{r=k}^{N-1}\binom{N-1}{r}}{(N-1)\binom{N-2}{k-1}}, \qquad \frac{2(N-2)}{Nk} \le d_k \le \frac{2}{k}. \tag{A.3a}$$

For the lower bound, after clearing the positive denominator it is enough to show

$$Nk\sum_{r=k}^{N-1}\binom{N-1}{r} \le 2^N(N-1)\binom{N-2}{k-1}. \tag{A.3b}$$

For $k \le N/2$, use the full binomial sum and $\binom{N-2}{k-1} \ge N-2$ away from the immediate endpoint $k=1$; for $k \ge 2, N \ge 4$, the remaining comparison follows from $Nk \le N^2/2 \le 2(N-1)(N-2)$. The case $k=1$ is immediate. For $k > N/2$, symmetry and monotonicity of the lower half of the binomial row give

$$\sum_{r=k}^{N-1}\binom{N-1}{r} \le (N-k)\frac{N-1}{k}\binom{N-2}{k-1},$$

and $N(N-k) \le 2^N$ completes the proof. The order $N=2$ is direct.

A.1 Standard column-imbalance directions

Let $\xi \in \mathbb{R}^{N+1}$ have coordinate sum zero and let $E(\xi)$ be the standard embedding in (4.8). By S_{N+1}-invariance and irreducibility of the standard module, it is enough to calculate the scalar for the canonical vector $\xi = e_x - e_y$; the resulting ratio then holds for every zero-sum ξ. Distinguish the labels x, y. The signed two-label quotient has positive channels

$$\begin{array}{r|l} P_k & v = x,\ x, y \notin B, \quad 1 \le k < N, \\ Q_k & v = x,\ y \in B, \quad 1 \le k \le N, \\ R_k & v \notin \{x, y\},\ x \in B,\ y \notin B, \quad 1 \le k < N, \end{array}$$

with the negative orientation obtained by exchanging x, y. Use the channel orders

$$\mathcal{G} = (P_1, \ldots, P_{N-1}, R_1, \ldots, R_{N-1}), \qquad \mathcal{Q} = (Q_1, \ldots, Q_N).$$

In these coordinates the operator is

$$H = \begin{pmatrix} S & C \\ -D & Q \end{pmatrix}, \tag{A.4s}$$

where all four displayed blocks are nonnegative. The nonzero signed rows are

$$P_k \mapsto \frac{k}{2N}P_k + \frac{N-k-1}{2N}P_{k+1} + \frac{1}{2N}Q_{k+1}, \tag{A.5s}$$

$$\begin{aligned} Q_k \mapsto \frac{k^2-1}{2kN}Q_k + \frac{N-k}{2N}Q_{k+1} - \frac{k-1}{2kN}P_{k-1} - \frac{N-k}{2kN}P_k \\ - \frac{(k-1)^2}{2kN}R_{k-1} - \frac{(k-1)(N-k)}{2kN}R_k, \end{aligned} \tag{A.6s}$$

and

$$\begin{aligned} R_k \mapsto \left\{\frac{k}{2N} + \frac{(k-1)(N-k)}{2kN}\right\}R_k + \frac{N-k-1}{2N}R_{k+1} + \frac{(k-1)^2}{2kN}R_{k-1} \\ + \frac{1}{2kN}Q_k + \frac{k-1}{2kN}P_{k-1} + \frac{N-k}{2kN}P_k. \end{aligned} \tag{A.7s}$$

Terms outside their physical ranges are absent. Define the binomial row

$$\sigma(R_k) = \frac{\binom{N-2}{k-1}}{2^{N-2}}, \qquad \sigma(P_k) = 0, \qquad \sigma(Q_k) = 0, \tag{A.8s}$$

and the physical reward

$$\begin{aligned}
\gamma(P_k) &= kd_k, \\
\gamma(R_k) &= Nd_k + \frac{(N+1)(k-1)}{k} d_{k-1}, \\
\gamma(Q_k) &= -q_k, \quad q_k = (N-k)d_k + \frac{(N+1)(k-1)}{k} d_{k-1} > 0.
\end{aligned} \tag{A.9s}$$

The standard-sector scalar is

$$\Phi_N = \sigma(I-H)^{-1}\gamma. \tag{A.10s}$$

This is the physical, rather than an auxiliary, reward. Indeed the first perturbation has feature coefficients

$$a_k = \frac{kd_k}{2(N+1)(N-1)}, \qquad b_k = \frac{Nd_k}{2(N+1)(N-1)} + \frac{(k-1)d_{k-1}}{2k(N-1)}, \tag{A.11s}$$

whose P, R, Q values are $a_k, b_k, a_k - b_k$. The averaged second perturbation annihilates the a feature and has b_k coefficient $\sigma(R_k)/[2(N+1)]$. Therefore

$$\boxed{\frac{\mathcal{R}_n^{(2)}(E(\xi))}{\|\xi\|^2} = \frac{\Phi_N}{4(N+1)^2(N-1)}.} \tag{A.12s}$$

It remains to prove $\Phi_N > 0$. Put

$$R_S = (I-S)^{-1}, \quad R_Q = (I-Q)^{-1}, \quad A = R_S C R_Q D,$$

where the displayed rows give $S\mathbf{1} \le (N-1)\mathbf{1}/N$ and $(Q\mathbf{1})_k = 1/2 - 1/(2kN)$. Thus both inverses are convergent Neumann series and are entrywise nonnegative. Also put

$$h = R_Q q, \qquad r_0 = \gamma_S - Ch, \qquad f_0 = R_S r_0. \tag{A.13s}$$

Schur elimination of the bad channel gives the exact alternating-phase identity

$$\Phi_N = \sigma(I+A)^{-1} f_0. \tag{A.14s}$$

For $N \ge 6$, define

$$W_1 = 2N^2 d_1, \quad W_2 = 2Nd_1, \quad W_k = \frac{4N(k-1)}{k} d_{k-1} \quad (3 \le k \le N). \tag{A.15s}$$

Then $(I-Q)W \ge q$. At ranks one and two the cleared residuals are $2(N^2 - N + 1)d_1$ and

$$\frac{3N-4}{2} d_1 - \frac{7(N-2)}{3} d_2 \ge \frac{2(N-2)(N-6)}{3N}.$$

For $k \ge 3$ the residual is $A_k d_{k-1} - B_k d_k$, with

$$A_k = \frac{(k-1)(3kN - 2k^2 - k + 2)}{k^2}, \qquad B_k = \frac{(3k+1)(N-k)}{k+1}.$$

Using (A.3a), it is at least $2P(N,k)/[Nk^2(k+1)]$. On setting $a = k-3$ and $m = N-k$,

$$\begin{aligned} P &= a^4 + a^3m + 10a^3 + 5a^2m + 37a^2 + 60a \\ &\quad + 2am(m-1) + 6m^2 - 22m + 32 > 0. \end{aligned} \tag{A.16s}$$

the last quadratic has discriminant -284. Thus $h \le W$. Direct substitution now gives

$$\begin{array}{c|c} P_1 & 0 \\ P_k & k(k-1)d_k/(k+1), \quad 2 \le k < N \\ R_1 & 0 \\ R_2 & Nd_2 + Nd_1/2 \\ R_k & Nd_k + \frac{k-1}{k}(N+1-2/k)d_{k-1}, \quad 3 \le k < N \end{array} \tag{A.17s}$$

for the coordinates of $\gamma_S - CW$. Hence $f_0 \ge 0$.

The bad exit and retention row sums satisfy

$$D\mathbf{1} = \frac{N-1}{2N}\mathbf{1}, \qquad (Q\mathbf{1})_k = \frac{1}{2} - \frac{1}{2kN}. \tag{A.18s}$$

Consequently $R_Q D\mathbf{1} \le \mathbf{1}$. With

$$z(P_k) = 1/N, \qquad z(R_k) = 2/(N+k),$$

one has $(I-S)z \ge C\mathbf{1}$. The P_k gap is $1/(2N^2)$; after $a = k-1, m = N-k$, the cleared R_k gap is

$$\begin{aligned} Z &= 4a^4 + 14a^3m + 18a^3 + 14a^2m^2 + 40a^2m + 30a^2 \\ &\quad + 4am^3 + 24am^2 + 36am + 20a + 3m^3 + 8m^2 + 9m + 4 > 0. \end{aligned} \tag{A.19s}$$

Applying the positive inverses proves

$$A\mathbf{1} \le \frac{2}{N+1}\mathbf{1}. \tag{A.20s}$$

Two elementary barriers control the first phase. The supersolution $u(P_k) = 4, u(R_k) = 8N$ gives $0 \le f_0 \le 8N\mathbf{1}$. Indeed its P residual is $2(N+1)/N \ge kd_k$, while its R residual exceeds the upper bound $2(2N+1)/k$ for $\gamma(R_k)$ by $2(4Nk - 4N + 1)/(Nk)$. Separately, $q_k \le 2N$ and (A.18s) imply $h \le 4N\mathbf{1}$. For $N \ge 7$, the subsolution $\ell(P_k) = 0, \ell(R_k) = 2N$ then gives

$$f_0(R_k) \ge 2N, \qquad \sigma f_0 \ge 2N. \tag{A.21s}$$

At $k = 1$ the margin is $2N - 6 - (N+1) \ge 0$; at $k \ge 2$ it is $[3N - 7 - 2k - 4/N]/k \ge 0$.

Set $c_N = 2/(N+1)$. Equations (A.20s)–(A.21s) and the Neumann series for $(I+A)^{-1}$ yield, for $N \ge 10$,

$$\Phi_N \ge 2N - 8N\sum_{j\ge1} c_N^j = \frac{2N(N-9)}{N-1} > 0. \tag{A.22s}$$

The remaining exact values are

(A.23s)

N	Φ_N	N	Φ_N
2	24/11	6	5758562957/248448224
3	261/40	7	141339691089527/4988552903680
4	343400/28657	8	15468663676289/466560376100
5	2268275/128288	9	19782952499295763/524622207176704.

This proves the standard-sector sign in every order. The supplementary verifier independently reconstructs the quotient, normalization, barriers, shifted polynomials, and every value in (A.23s).

A.2 Antisymmetric balanced directions

Let $\delta = -\delta^{\mathsf{T}}$ have zero row sums and set

$$x(B, v) = \sum_{i \in B} \delta_{vi}. \tag{A.4}$$

Direct expansion of the active moves gives

$$\Delta h(B, v) = \frac{d_k}{2} x(B, v), \qquad k = |B|. \tag{A.5}$$

The Poisson gradients satisfy

$$d_1 > d_2 > \cdots > d_{N-1} > 0. \tag{A.6}$$

To see this, couple adjacent complete rank chains by refreshing the same bit. Until their distinguished unequal bit is refreshed, the forcing difference is $1/(X+1) - 1/(X+2) > 0$; adding a shared occupied bit decreases that difference. Summing the convergent Poisson series proves positivity and strict monotonicity.

For any rank sequence f_k,

$$K_0\{f_k x(B, v)\} = \frac{k f_k + (N-k-1) f_{k+1}}{2N} x(B, v). \tag{A.7}$$

Thus $G\Delta h = r_k x$, where $r_N = 0$ and

$$(2N - k) r_k - (N - k - 1) r_{k+1} = N d_k. \tag{A.8}$$

Backward induction using (A.6) gives

$$0 \le r_{k+1} < r_k \le \frac{N}{N+1} d_k. \tag{A.9}$$

Put $T = \|\delta\|_F^2$. Sampling without replacement and applying Δ a second time yields

$$\begin{aligned}\mathbb{E}_k[\Delta(r_k x)] = \frac{T}{2nN}\Big[&\frac{k(N-k)}{N-1}(r_k - r_{k+1}) + (N-k) r_{k+1} \\ &+ \frac{(k-1)(N-k+1)}{N-1}(r_{k-1} - r_k) + (N-k+1) r_k\Big].\end{aligned} \tag{A.10}$$

The coefficient of the absent $r_0 - r_1$ term is zero. Every displayed term is nonnegative by (A.9), and the total is strict when $T > 0$. Averaging with (A.1) proves positivity of the antisymmetric sector for every $N \ge 2$.

A.3 Symmetric balanced directions

For a symmetric balanced δ, put

$$x(B, v) = \sum_{i \in B} \delta_{vi}, \qquad z(B) = \sum_{w \neq i \in B} \delta_{wi}. \tag{A.11}$$

Functions in this sector have the form $a_k x + b_k z$. Transpose the exact two-channel rank operator and split its positive and negative channels:

$$H = K^{\mathsf{T}} = \begin{pmatrix} S & C \\ -D & Q \end{pmatrix}. \tag{A.12}$$

Every displayed block is entrywise nonnegative. The nonzero entries are

$$\begin{aligned} S_{k,k} &= \frac{k}{2N}, & S_{k,k-1} &= \frac{N-k}{2N}, \\ Q_{k,k} &= \frac{N(k-2)+k}{2kN}, & Q_{k,k-1} &= \frac{N-k-1}{2N}, \\ Q_{k,k+1} &= \frac{k(k-1)}{2(k+1)N}, & D_{k,k-1} &= \frac{1}{N}, \end{aligned} \tag{A.13}$$

and

$$C_{k-1,k} = \frac{k-1}{2kN}, \qquad C_{k,k} = \frac{N-k}{2kN}. \tag{A.14}$$

The a channel is indexed by $1 \le k < N$ and the b channel by $2 \le k < N$; entries outside these ranges are zero. The positive source is

$$s_k^a = \frac{d_k}{2}, \qquad s_k^b = \frac{d_{k-1}}{2k}, \tag{A.15}$$

and the good/bad reward is

$$g_k^a = \frac{\binom{N-2}{k-1}}{2^{N-1}(N+1)}, \qquad g_k^b = -\frac{\binom{N-3}{k-2}}{2^{N-2}(N+1)}. \tag{A.16}$$

The normalization connecting this quotient to the physical Hessian is exact, not merely sign preserving. If $(a,b) = (I-K)^{-1}s$, set $b_1 = a_N = b_N = 0$ and

$$r_k = a_k - \frac{2(k-1)}{N-2} b_k,$$

then the fixed-count orbit identities give

$$\frac{\mathcal{R}_n^{(2)}(\delta)}{\|\delta\|_F^2} = \sum_{k=1}^{N-1} \pi_k \frac{N-k}{(N-1)(N+1)} r_k = g^{\mathsf{T}}(I-K)^{-1}s =: \mathcal{S}_N. \tag{A.17}$$

Here is the all-order labelled normalization. Let $c = 1/[nN2^{N-1}]$ and write $\mu = \nu_0\Delta$. For an output state (B,v) of rank k, the four predecessor families contribute, respectively,

$$\frac{c}{2}kx(B,v), \quad \frac{c}{2}(k-1)x(B,v), \quad \frac{c}{2}(N-k)x(B,v), \quad \frac{c}{2}(N-k+1)x(B,v).$$

Thus their sum is the exact labelled current $\mu(B,v) = x(B,v)/(n2^{N-1})$, including the boundary ranks. Define $f(B,v) = a_{|B|}x(B,v) + b_{|B|}z(B)$. Conditional on a uniform rank-k orbit, sampling without replacement gives, with $R_v = \sum_i \delta_{vi}^2$,

$$\mathbb{E}[x^2 \mid v] = \frac{k(N-k)}{N(N-1)} R_v, \qquad \mathbb{E}[xz \mid v] = -\frac{2k(k-1)(N-k)}{N(N-1)(N-2)} R_v. \tag{A.17a}$$

For the cross term, separate the two coincidences in the ordered triple from the all-distinct case; the row and column sums reduce the resulting full sums to $-2R_v$. Summing R_v over v gives $\|\delta\|_F^2$. Therefore

$$\frac{\nu_0\Delta f}{\|\delta\|_F^2} = \sum_{k=1}^{N-1} \pi_k \frac{N-k}{(N-1)(N+1)} \left(a_k - \frac{2(k-1)}{N-2} b_k \right), \tag{A.17b}$$

which is the first expression in (A.17). The identities

$$\pi_k \frac{N-k}{N-1} = \frac{\binom{N-2}{k-1}}{2^{N-1}}, \qquad \frac{k-1}{N-2}\binom{N-2}{k-1} = \binom{N-3}{k-2}$$

show that the coefficients in (A.17b) are exactly g_k^a, g_k^b. Hence the orbit average is $g^{\mathsf{T}}(a,b) = g^{\mathsf{T}}(I-K)^{-1}s$. This is an all-order labelled derivation; the exact replay independently reconstructs it for $4 \le n \le 12$ before taking a sign.

Put $R_S = (I-S)^{-1}$ and $R_Q = (I-Q)^{-1}$. Here S has row sums at most $1/2$, while the displayed tridiagonal Q is substochastic and every communicating class reaches a strict boundary row. Hence both inverses are convergent entrywise-nonnegative Neumann series. Define

$$\begin{aligned} W &= R_Q(-g^b), & r &= g^a - CW, & f_0 &= R_S r, \\ A &= R_S C R_Q D, & \ell &= (s^a)^{\mathsf{T}} - (s^b)^{\mathsf{T}} R_Q D, & \mathcal{D} &= (s^b)^{\mathsf{T}} W. \end{aligned} \tag{A.18}$$

Solving the bad equation first gives the exact Schur identity

$$\mathcal{S}_N = \ell(I+A)^{-1} f_0 - \mathcal{D}. \tag{A.19}$$

One factor of the nonnegative matrix A is one completed visit to the bad channel. The inverse in (A.19) therefore alternates successive re-entries, so a first-excursion estimate alone would not suffice.

We now give the full phase certificate. Put

$$v = R_S g^a, \qquad v_k = t_k g_k^a, \qquad t_1 = \frac{2N}{2N-1}, \quad t_k = \frac{2N + (k-1)t_{k-1}}{2N-k}. \tag{A.20}$$

Induction gives

$$t_k \le \frac{N^2-k}{(N-2)(N-k)}. \tag{A.21}$$

The base difference is $(5N-1)/[(N-2)(2N-1)]$; after one step and the substitution $m = N-k$, the remaining numerator is $N^3 - N^2 + 2Nm^2 + 2Nm + 2m^3 + 3m^2 + m > 0$.

For $N \ge 24$, let $\overline{W} = (7N/25)(-g^b)$. Direct substitution shows

$$(I-Q)\overline{W} \ge -g^b, \qquad C\overline{W} \le \frac{14}{25} g^a. \tag{A.22}$$

For the first inequality, after clearing positive binomial factors the interior residual is

$$P_N(k) = 21N^2k + 14N^2 - 64Nk^2 - 57Nk + 50k^3 + 36k^2 - 14k. \tag{A.23}$$

For $N \ge 25$, $\operatorname{disc}_k P_N = -8B_N$, where the coefficients of B_{25+M} are

$$(5733, 736211, 37934833, 982793213, 12893444604, 70866894144, 41021531998). \tag{A.24}$$

Thus P_N has one real, negative root and is positive for $k \ge 0$; at $N = 24$ its exact minimum over the physical ranks is 24. Applying R_Q in (A.22) yields

$$\frac{11}{25} v \le f_0 \le v. \tag{A.25}$$

For $2 \le k < N$, define

$$\widehat{h}_k = \frac{k}{N-2} g_{k-1}^a. \tag{A.26}$$

Equations (A.13) and (A.21) give $(I-Q)\widehat{h} \ge Dv$ and

$$C\widehat{h} \le c_N g^a, \qquad c_N = \frac{2N-5}{2N(N-2)}.$$

Indeed, the interior ratio in the first comparison is $(N^2-k+1)/[N(N-2)(N-k+1)]$, and the largest row ratio in the second is at rank $N-2$. At $k=2$ the first boundary residual is $(N+1)/[N(N-2)]$, equal to the interior formula; at $k=N-1$ it is $(N^2+1)/[2N(N-2)]$, which exceeds the interior value $(N^2-N+2)/[2N(N-2)]$. Thus (A.21) dominates every boundary row as well. Consequently

$$Av \le c_N v. \tag{A.27}$$

For the left occupation debt, put

$$Y_k = \frac{2(k+1)}{3k(k-1)} \qquad (2 \le k < N). \tag{A.28}$$

With $B = Q^\mathsf{T}$, an interior substitution gives

$$[(I-B)Y]_k - \frac{1}{k(k-1)} = \frac{4Nk + 2N + 2k^3 - k^2 - 5k}{3Nk^2(k-1)(k+1)} > 0, \tag{A.29}$$

while the boundary residuals at $k=2$ and $k=N-1$ are, respectively, $(5N+7)/(18N) > 0$ and $(N^2-N+2)/[3N(N-2)(N-1)^2] > 0$. Since $s^b_k \le 1/[k(k-1)]$, this proves $(s^b)^\mathsf{T} R_Q \le Y^\mathsf{T}$ and hence $\ell \ge \overline{\ell} > 0$, where

$$\overline{\ell}_j = \frac{3Nj + 3N - 8j - 10}{3Nj(j+1)}. \tag{A.30}$$

The absent bad state above the top rank only increases the final margin.

The bad debt and first phase now satisfy

$$\mathcal{D} \le \sum_{j=1}^{N-2} \frac{4(j+2)}{3(j+1)(N-2)} g^a_j, \qquad \ell f_0 \ge \frac{11}{25} \sum_{j=1}^{N-1} \overline{\ell}_j t_j g^a_j. \tag{A.31}$$

Define

$$\begin{aligned} \beta_N &= \max_{1 \le j \le N-2} \frac{4(j+2)}{3(j+1)(N-2)} \left(\frac{11}{25} \overline{\ell}_j t_j \right)^{-1}, \\ \varepsilon_N &= \frac{25}{11} \frac{c_N}{1-c_N}. \end{aligned} \tag{A.32}$$

Exact rational recurrence (A.20) gives $\beta_N + \varepsilon_N < 1$ for $40 \le N \le 287$; its smallest margin is

$$1 - \beta_{40} - \varepsilon_{40} = \frac{639304267467075678841}{1153695882967924671447\!16} > 0. \tag{A.33}$$

This is a finite exact calculation.

For $N \ge 288$, induction in (A.20) gives $t_j \ge N/(N-j+\sqrt{N/2})$. Indeed, substituting the inductive bound leaves the numerator $2u(N-j+u) + u - N$, where $u = \sqrt{N/2}$; it is minimized at $j = N-1$ and equals $3u > 0$. Since $u \le N/24$ in this range, $t_j \ge 24N/(25N-24j)$. The bound $\beta_N \le 19/20$ is then equivalent, after clearing positive factors, to $G_N(j) > 0$, where

$$\begin{aligned} G_N(j) = {} & 1881N^2 j + 1881N^2 - 6250Nj^2 - 21278Nj - 10032N \\ & + 6000j^3 + 12000j^2 + 10032j + 12540. \end{aligned} \tag{A.34}$$

Its discriminant is $-500D_N$, and the coefficients of D_{288+M} are

$$\begin{gathered} 43034652243, 71940715263252, 50075984929826764, \\ 18577025353519361184, 3873653773133773223808, \\ 430447522448513182675968, 19913301794000751100222464. \end{gathered} \tag{A.35}$$

Thus G_N has one negative real root and is positive on every physical rank. Also $\varepsilon_N \le 1/20$ for $N \ge 46$. Therefore $\beta_N + \varepsilon_N < 1$ for all $N \ge 288$.

Finally, (A.25), (A.27), and positivity of ℓ give

$$\left|\ell(I+A)^{-1}f_0 - \ell f_0\right| \le \varepsilon_N \ell f_0.$$

Together with (A.31) this yields

$$\mathcal{S}_N \ge (1-\beta_N-\varepsilon_N)\ell f_0 > 0 \tag{A.36}$$

for every $N \ge 40$. The finitely many orders $3 \le N \le 39$ are solved exactly over $\mathbb{Q}$; the first values are $\mathcal{S}_3 = 3/208$ and $\mathcal{S}_4 = 359/26660$.

The supplementary verifier reconstructs every rational boundary solve, cleared polynomial, shifted coefficient list, and the labelled normalizations (A.12s) and (A.17). Combining the three subsections proves Theorem 8.

B Two symmetric weighted K_4 families

Let $G_{13}(x)$ have one core vertex, three satellites, unit core–satellite weights, and satellite–satellite weight $x > 0$. Let $G_{22}(x, y)$ have two vertex pairs, internal-pair weights $x, y > 0$, and four unit cross-pair weights. These are the full $S_1 \times S_3$ and $S_2 \times S_2$ invariant families after common scaling.

Figure 2: The two invariant weighted K_4 families. Unlabeled edges have unit weight.

A two-class graph with class sizes p, q, internal weights α, β, and cross weight γ is strongly lumpable by mutant counts (i, j). The four state-changing probabilities follow directly from (2.2):

$$\begin{aligned}
P_{i,j}^{i+1,j} &= \frac{p-i}{p+q}\,\frac{r(i\alpha+j\gamma)}{r(i\alpha+j\gamma)+(p-i-1)\alpha+(q-j)\gamma}, \\
P_{i,j}^{i-1,j} &= \frac{i}{p+q}\,\frac{(p-i)\alpha+(q-j)\gamma}{r((i-1)\alpha+j\gamma)+(p-i)\alpha+(q-j)\gamma}, \\
P_{i,j}^{i,j+1} &= \frac{q-j}{p+q}\,\frac{r(j\beta+i\gamma)}{r(j\beta+i\gamma)+(q-j-1)\beta+(p-i)\gamma}, \\
P_{i,j}^{i,j-1} &= \frac{j}{p+q}\,\frac{(q-j)\beta+(p-i)\gamma}{r((j-1)\beta+i\gamma)+(q-j)\beta+(p-i)\gamma}.
\end{aligned} \tag{B.1}$$

For the $1+3$ family, exact elimination gives

$$\rho_{\mathrm{dB}}(G_{13}(x),r)-\rho_{\mathrm{dB}}(J_4,r)=-\frac{3r^2(r-1)(x-1)^2F_{13}}{4(r^2+r+1)P_{13}}, \tag{B.2}$$

where

$$\begin{aligned}F_{13}&=2r^4x^2+2r^4x+11r^3x^2+14r^3x+3r^3+21r^2x^2+29r^2x+12r^2\\&\quad+16rx^2+22rx+8r+4x^2+5x+1,\end{aligned}$$

and

$$\begin{aligned}P_{13}&=8x^2(x+1)(r^6+1)+(6x^4+36x^3+46x^2+16x)(r^5+r)\\&\quad+(27x^4+73x^3+85x^2+59x+12)(r^4+r^2)\\&\quad+(42x^4+90x^3+106x^2+66x+24)r^3.\end{aligned}$$

Every coefficient is positive.

For the $2+2$ family,

$$\rho_{\mathrm{dB}}(G_{22}(x,y),r)-\rho_{\mathrm{dB}}(J_4,r)=-\frac{r^2(r-1)H_{22}(x,y,r)}{4(r^2+r+1)P_{22}(x,y,r)}. \tag{B.3}$$

The reduced P_{22} has 123 positive integer monomials. Independently, the holding-step-free transient matrix M_{22} satisfies

$$\det M_{22}=\frac{P_{22}}{128L_{22}}, \tag{B.4}$$

where

$$\begin{aligned}L_{22}&=(2r+x)(2r+y)(rx+2)(ry+2)(r+x+1)(r+y+1)\\&\quad\times(rx+r+1)(ry+r+1)>0.\end{aligned}$$

Absorption makes M_{22} a nonsingular M-matrix, so $P_{22}>0$.

Set $g=\sqrt{xy}>0$, $d=(\sqrt{x}-\sqrt{y})^2\geq 0$, and $t=r-1>0$. Then

$$H_{22}=C_0(g,t)+C_1(g,t)d+C_2(g,t)d^2+C_3(g,t)d^3+C_4(g,t)d^4. \tag{B.5}$$

Here

$$C_0=2t(g-1)^2(g+1)(t+1)R_0,$$

$$\begin{aligned}R_0&=(2g^2+2g)t^4+(g^3+10g^2+21g+6)t^3\\&\quad+(3g^3+26g^2+61g+38)t^2+(4g^3+32g^2+80g+64)t\\&\quad+2g^3+16g^2+40g+32,\end{aligned}$$

and

$$\begin{aligned}
C_1 &= 2(g^4+4g^3-2g^2+4g+1)t^6+(11g^4+108g^3+76g^2+60g+17)t^5\\
&\quad+2(40g^4+288g^3+393g^2+176g+19)t^4\\
&\quad+(16g^5+331g^4+1704g^3+2766g^2+1360g+39)t^3\\
&\quad+2(28g^5+337g^4+1426g^3+2399g^2+1378g+12)t^2\\
&\quad+2(g+2)(32g^4+263g^3+722g^2+661g+2)t\\
&\quad+24g(g+2)(g+4)(g^2+4g+5),\\
C_2 &= 2(g^2+1)t^6+3(9g^2+26g+5)t^5+2(18g^3+120g^2+321g+44)t^4\\
&\quad+2(2g^4+105g^3+525g^2+1071g+170)t^3\\
&\quad+(14g^4+474g^3+2201g^2+3648g+689)t^2\\
&\quad+2(8g^4+240g^3+1080g^2+1587g+331)t\\
&\quad+6(g^4+30g^3+133g^2+186g+40),\\
C_3 &= 13t^5+(6g^2+48g+107)t^4+(35g^2+312g+357)t^3\\
&\quad+(79g^2+744g+608)t^2+(80g^2+768g+529)t+6(5g^2+48g+31),\\
C_4 &= 6t^4+39t^3+93t^2+96t+36.
\end{aligned}$$

Every displayed coefficient is positive except for the single grouped factor

$$g^4+4g^3-2g^2+4g+1=(g^2-1)^2+4g(g^2+1)>0. \tag{B.6}$$

Thus $C_j>0$ for $j\ge 1$ and $C_0\ge 0$. If $d>0$, then $C_1 d>0$; if $d=0$, $H_{22}=C_0$ vanishes only at $g=1$.

Theorem 12 (symmetric weighted K_4 families)**.** *For every $r>1$,*

$$\rho_{\mathrm{dB}}(G_{13}(x),r)\le\rho_{\mathrm{dB}}(J_4,r),\qquad \rho_{\mathrm{dB}}(G_{22}(x,y),r)\le\rho_{\mathrm{dB}}(J_4,r).$$

Equality holds only at $x=1$ in the first family and $x=y=1$ in the second.

This is not a classification of the unrestricted six-edge weighted K_4.

C Quantifier and scope ledger

The statements in this paper have deliberately different orders of quantification.

Fitness-two local theorem. For every $n\ge 3$ there is an $\eta_n>0$ such that every positive loopless row-stochastic P with $0<\|P-J_n\|<\eta_n$ satisfies $\rho_{\mathrm{dB}}(P,2)<\rho_{\mathrm{dB}}(J_n,2)$.

Full-tangent curvature. For every $n\ge 3$ and every $\delta\in\mathcal{T}_n\setminus\{0\}$,

$$D\rho_{\mathrm{dB}}(J_n,2)[\delta]=0,\qquad D^2\rho_{\mathrm{dB}}(J_n,2)[\delta,\delta]<0.$$

Fixed-graph strong selection. For every fixed P that is not dynamically complete, there is an $r_0(P)$ such that $\rho_{\mathrm{dB}}(P,r)<\rho_{\mathrm{dB}}(J_n,r)$ for every $r>r_0(P)$.

Triangle theorem. Every positive weighted triangle and every $r>1$ obey complete-graph maximality, with strictness away from equal weights.

Open global fitness-two statement. It remains open whether every n and every undirected weighted P satisfy $\rho_{\mathrm{dB}}(P, 2) \leq \rho_{\mathrm{dB}}(J_n, 2)$.

Open asymptotic-family statement. It also remains open whether there is one sequence (P_n) such that, for every fixed $r > 1$, some $n_0(r)$ makes every P_n with $n \geq n_0(r)$ an amplifier.

The proved rows do not imply either open row. In particular, local optimality has a radius η_n that need not be bounded below, and the strong-selection threshold $r_0(P_n)$ may diverge with n.

References

B. Allen, G. Lippner, Y.-T. Chen, B. Fotouhi, N. Momeni, S.-T. Yau, and M. A. Nowak. Evolutionary dynamics on any population structure. *Nature* **544** (2017), 227–230. doi:10.1038/nature21723.

B. Allen, C. Sample, R. Jencks, J. Withers, P. Steinhagen, L. Brizuela, J. Kolodny, D. Parke, G. Lippner, and Y. A. Dementieva. Transient amplifiers of selection and reducers of fixation for death–Birth updating on graphs. *PLoS Computational Biology* **16** (2020), e1007529. doi:10.1371/journal.pcbi.1007529.

T. E. Harris. Additive set-valued Markov processes and graphical methods. *The Annals of Probability* **6** (1978), 355–378. doi:10.1214/aop/1176995523.

L. Hindersin and A. Traulsen. Most undirected random graphs are amplifiers of selection for birth–death dynamics, but suppressors of selection for death–birth dynamics. *PLoS Computational Biology* **11** (2015), e1004437. doi:10.1371/journal.pcbi.1004437.

K. Kaveh, N. L. Komarova, and M. Kohandel. The duality of spatial death–birth and birth–death processes and limitations of the isothermal theorem. *Royal Society Open Science* **2** (2015), 140465. doi:10.1098/rsos.140465.

A. Kriebel. No universal death–Birth amplifier on a finite weighted population structure. Unrefereed research release, version 1.0.0 (2026). doi:10.5281/zenodo.21753405.

E. Lieberman, C. Hauert, and M. A. Nowak. Evolutionary dynamics on graphs. *Nature* **433** (2005), 312–316. doi:10.1038/nature03204.

N. Masuda and H. Ohtsuki. Evolutionary dynamics and fixation probabilities in directed networks. *New Journal of Physics* **11** (2009), 033012. doi:10.1088/1367-2630/11/3/033012.

P. A. P. Moran. Random processes in genetics. *Mathematical Proceedings of the Cambridge Philosophical Society* **54** (1958), 60–71. doi:10.1017/S0305004100033193.

P. Shakarian, P. Roos, and A. Johnson. A review of evolutionary graph theory with applications to game theory. *BioSystems* **107** (2012), 66–80. doi:10.1016/j.biosystems.2011.09.006.

J. Tkadlec, A. Pavlogiannis, K. Chatterjee, and M. A. Nowak. Limits on amplifiers of natural selection under death–Birth updating. *PLoS Computational Biology* **16** (2020), e1007494. doi:10.1371/journal.pcbi.1007494.